

P-8

PREBOARD – 1
ARTIFICIAL INTELLIGENCE (CODE - 417)
CLASS X (SESSION 2024-2025)

Max. Time: 2 Hours**Max. Marks: 50****General Instructions:**

1. Please read the instructions carefully.
2. This Question Paper consists of **21 questions** in two sections: Section A & Section B.
3. Section A has Objective type questions whereas Section B contains Subjective type questions.
4. **Out of the given (5 + 16 =) 21 questions, a candidate has to answer (5 + 10 =) 15 questions in the allotted (maximum) time of 2 hours.**
5. All questions of a particular section must be attempted in the correct order.
6. **SECTION A - OBJECTIVE TYPE QUESTIONS (24 MARKS):**
 - i. This section has 05 questions.
 - ii. Marks allotted are mentioned against each question/part.
 - iii. There is no negative marking.
 - iv. Do as per the instructions given.
7. **SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):**
 - i. This section has 16 questions.
 - ii. A candidate has to do 10 questions.
 - iii. Do as per the instructions given.
 - iv. Marks allotted are mentioned against each question/part.

SECTION-A		
Q. 1	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)	
i.	<i>"M D Gulati started with a small shop with his focus, dedication and clear ideas, MDH became one of the most popular brands in India besides having a good reputation all over the world".</i> Which self-management skill is clearly visible in the given statement?	1
ii.	When you bring the mouse over a file in File Explorer, it will show the details of that file. This is known as (a) Drag and drop (b) Double click (c) Hover (d) Single click	1
iii.	Assertion(A): A doctor works for a renowned hospital. Reason(R): The statement given above is an example of wage employment. (a) Both A and R are correct and R is the correct explanation of A (b) Both A and R are correct but R is NOT the correct explanation of A (c) A is correct but R is not correct (d) A is not correct but R is correct	1
iv.	_____ the work is all about identifying and noting how we spent our time, and analysing how to spend our time effectively. (a) Organising (b) Prioritising (c) Controlling (d) Tracking	1
v.	Remya traveled to Sweden from India to pursue her higher education. But she doesn't know how to speak Swedish (language of Sweden). Because of this, she was unable to find a part time job. This is an example of _____ (a) Interpersonal barrier (b) Physical barrier (c) Organisational barrier	1

	(d) Linguistic barrier	
vi.	“Efforts are made to increase the solar power generation so that our electricity needs are met and at the same time we do not pollute the environment or use up natural resources”. Which SDG can you relate this statement to? (a) Life on land (b) Clean water and sanitation (c) Affordable and clean energy (d) Reduced inequalities	1
SECTION-B		
Q. 2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i	Assertion (A): One person may excel as a singer while another excels as an athlete. Reason (R): Humans have different types of intelligences that vary in levels among individuals. (a) Both A and R are correct, and R is the correct explanation of A (b) Both A and R are correct, but R is not the correct explanation of A (c) A is correct, but R is incorrect (d) A is incorrect, but R is correct	1
ii	A company was fined for collecting personal data from users without their consent and using it for targeted advertising. Which term best describes the issue in this scenario? (a) Data Security (b) AI Ethics (c) Data Privacy (d) AI Transparency	1
iii	Statement 1: Machine learning models improve their performance based on the data they are trained on. Statement 2: The training dataset is used to evaluate the model's performance. (a) Both Statement 1 and Statement 2 are correct (b) Both Statement 1 and Statement 2 are incorrect (c) Statement 1 is correct but Statement 2 is incorrect (d) Statement 2 is correct but Statement 1 is incorrect	1

iv	_____ is a common file format used to store tabular data in rows and columns. (a) txt (b) pdf (c) csv (d) gif	1
v	A corpus contains 5 documents, and the word 'technology' appears twice in document 3. How would you categorize the word 'technology'? (a) Stop word (b) Rare word (c) Frequent word (d) Removable word	1
vi	In a corpus of 6 documents, the word 'innovation' appears only once in document 2. How would you classify the word 'innovation'? (a) Stop word (b) Rare word (c) Frequent word (d) Removable word	1
Q 3	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i	Read the examples given below: i. Using a voice assistant to set reminders ii. GPS navigation in mobile devices iii. Automated coffee machine that brews coffee at a scheduled time iv. Basic calculator performing arithmetic operations Choose the options that are not AI: (a) i and ii (b) iii and iv (c) ii and iii (d) i, iii and iv	1
ii	_____ describes the characteristics or properties of the data we aim to analyze.	1
iii	Which of the following is an application of machine learning? (a) Speech recognition (b) Word processing	1

	(c) Image viewing (d) Basic text editing	
iv	_____ is the technique used to assign a label to an entire image based on its content. (a) Object detection (b) Image segmentation (c) Classification (d) Instance segmentation	1
v	Identify the given AI system type: It analyzes data patterns and makes predictions based on historical data. It continually improves its accuracy through feedback and adapts to new information over time. (a) Rule-based system (b) Expert system (c) Machine learning model (d) Neural network	1
vi	The ROC-AUC score evaluates the trade-off between: (a) Precision and Recall (b) True Positive Rate and False Positive Rate (c) Accuracy and Precision (d) Sensitivity and Specificity	1
Q 4	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i	Rahul studied various programming concepts but struggled to remember the term that describes a programming paradigm based on the concept of "objects" that can contain data and code. Can you help him with the correct term?	1
ii	Statement 1: In supervised learning, the model is trained using labeled data. Statement 2: Unsupervised learning requires labeled data for training the model. (a) Both Statement 1 and Statement 2 are correct (b) Both Statement 1 and Statement 2 are incorrect (c) Statement 1 is correct but Statement 2 is incorrect (d) Statement 2 is correct but Statement 1 is incorrect	1
iii	Aisha developed a model to forecast sales based on various factors such as seasonality, marketing spend, and customer demographics. She ensured her model had high accuracy and a good F1 score. Which of the statements below is	1

	incorrect? (a) The data collected on seasonality, marketing spend, and customer demographics is known as Training Data. (b) The data used to evaluate the model's performance is called Validation Data. (c) Training data is usually larger than validation data. (d) Data collection for model training and validation occurs during the Data Preparation phase.	
iv	How many dimensions are typically used to represent a grayscale image?	1
v	Which feature of NLP helps in understanding the emotions of the people mentioned with the feedback? (a) Virtual Assistants (b) Sentiment Analysis (c) Text classification (d) Automatic Summarization	1
vi	Maya developed a spam email classifier using a dataset that contained labeled emails for training. However, when she tested the model using the same dataset, it achieved perfect accuracy but failed to classify new, unseen emails correctly. What is this concept called?	1
Q.5	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i	Aditi, a student of class XII developed a chatbot that clarifies the doubts of Economics students. She trained the software with lots of data sets catering to all difficulty levels. If any student would type or ask questions related to Economics, the software would give an instant reply. Identify the domain of AI in the given scenario. (a) Computer Vision (b) Data Science (c) Natural Language Processing (d) None of these	1
ii	Which evaluation metric calculates the proportion of correctly classified instances over the total number of instances?	1
iii	_____ refers to a model's ability to correctly identify positive instances. (a) Recall (b) Precision (c) Accuracy (d) F1-score	1
iv	What does the term "algorithm" refer to in the context of artificial intelligence? (a) A set of data (b) A process or set of rules for solving a problem	1

	(c) A programming language (d) A type of hardware	
V	Which technique represents text data as a collection of unique words and their frequency in a document? (a) Tokenization (b) Sentence Segmentation (c) Bag of Words (d) Text Normalization make a question similar to this in AI according to class 10 ncert AI syllabus	1
Vi	In which of the following scenarios would a high false negative cost be more concerning? (a) Spam detection (b) Fraud detection (c) Movie recommendation (d) Sentiment analysis	1
SECTION-B: SUBJECTIVE TYPE QUESTIONS Answer any 3 out of the given 5 questions on employability skills Answer each question in 20-30 words. (2X3=6marks)		
Q 6	List two best practices for training an effective machine learning model.	2
Q 7	What is the importance of defining a clear problem statement in an AI project?	2
Q 8	Explain the difference between supervised and unsupervised learning in artificial intelligence.	2
Q 9	How is artificial intelligence helping businesses improve their operations?	2
Q 10	Mention any two ways in which artificial intelligence can contribute to data analysis in various fields.	2
Answer any 4 out of the given 6 questions in 20-30 words each (2 x 4 =8 marks)		
Q 11	Why do AI systems require large amounts of data for training and improving their performance?	2
Q 12	Sirisha and Divisha want to make a model which will organize the unlabeled input data into groups based on features. Which learning model should they use and why?	2
Q 13	Ajay wants to access data from various sources. Suggest him any two points that he needs to keep in mind while accessing data from any data source.	2

Q 14	Explain the term "feature extraction" in the context of machine learning, and provide an example.	2
Q 15	Identify any two key features in the given dataset and explain their importance in building a machine learning model. Dataset: Customer information for an online retail store, including age, purchase history, and feedback scores.	2
Q 16	Draw the confusion matrix for the following data • the number of true positive = 100 • the number of true negative 47 • the number of false positive = 62 • the number of false negative = 290	2
Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)		
Q 17	Your younger sibling is curious about how chatbots work. Explain the term "chatbot" and provide an example of how a chatbot can assist users.	4
Q 18	Riya wants to understand the steps involved in building a machine learning model. Explain to her the following: (a) The data collection process (b) The model evaluation metrics used to assess the model's performance.	4
Q 19	Rahul is interested in the lifecycle of an AI project. Explain to him the following: (a) The importance of data preprocessing (b) The role of model training in the development of AI applications.	4
Q 20	We can make decisions based on emotions and experiences, but AI systems operate differently. What challenges do you think AI might face when trying to replicate human decision-making? Explain.	4
Q 21	Humans have the ability to learn from context and experiences, but AI relies on data. What challenges do you think AI might encounter when trying to understand context in natural language? Explain.	4
